Periodic infinite ringsets in the Lipschitz quaternions

Carptopus
carptopus@163.com

23 August 2026

Public Beta v0.1. The complete manuscript has passed independent proof and manuscript-level zero-trust audits. External mathematical review and journal peer review are pending.

Abstract

Let $\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}]$ be the ring of Lipschitz quaternions. Given a periodic word $w : \mathbb{Z} \to \{\mathbf{i}, \mathbf{j}\}$, we study the infinite set

$$S_w = \{a + w(a) : a \in \mathbb{Z}\} \subset \mathbf{L}.$$

A subset $S \subseteq \mathbf{L}$ is a ringset if the polynomials over the rational quaternion algebra that are integer-valued on S form a ring. We give a complete criterion when a prescribed period m is odd and squarefree. The set S_w is a ringset if and only if w is nonconstant and, for every prime $p \mid m$ with $p \equiv 3 \pmod 4$, the restriction of w to each residue class modulo p is nonconstant. Consequently, if every prime divisor of m is congruent to 1 modulo 4, every nonconstant binary word of period m produces an infinite ringset.

The proof combines the residue-null-ideal criterion for noncommutative integer-valued polynomials with two local mechanisms. At split prime powers, rank-one idempotents in $\mathbf{L}/p^e\mathbf{L} \simeq M_2(\mathbb{Z}/p^e\mathbb{Z})$ show that one chosen letter above each real residue already forces a core set. At primes congruent to 3 modulo 4, a Frobenius real-part extractor gives an explicit right-multiplication obstruction whenever a residue fibre has only one letter. Periodic congruence orbits and the Chinese remainder theorem then give the global classification.

1 Introduction

Let $\mathbb{D}$ denote the rational Hamilton quaternion algebra and let

$$\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}], \qquad \mathbf{i}^2 = \mathbf{j}^2 = -1, \qquad \mathbf{ij} = \mathbf{k} = -\mathbf{ji}.$$

The variable in $\mathbb{D}[x]$ is central, and polynomials are evaluated on the right: if $f(x) = \sum_d c_d x^d$, then $f(\alpha) = \sum_d c_d \alpha^d$. For $S \subseteq \mathbf{L}$, put

$$\mathrm{Int}(S, \mathbf{L}) = \{f \in \mathbb{D}[x] : f(S) \subseteq \mathbf{L}\}.$$

This set is always a left $\mathbf{L}$-module, but in the noncommutative setting it need not be closed under polynomial multiplication. Following Werner [4], S is called a *ringset* if $\mathrm{Int}(S, \mathbf{L})$ is a subring of $\mathbb{D}[x]$.

Werner [4] completely classified the finite ringsets in the Lipschitz and Hurwitz quaternion rings. The infinite problem behaves differently. In particular, a finite set can be decomposed into its minimal-polynomial classes, whereas this reduction fails for infinite sets. Werner's Proposition 5.6 gives a sufficient congruence-pairing condition and Example 5.9 constructs an infinite ringset whose individual minimal-polynomial fibres are all non-ringsets.

We examine the simplest stationary version of that phenomenon. Let $m \in \mathbb{Z}_{\geq 1}$ and let $w : \mathbb{Z} \to \{\mathbf{i}, \mathbf{j}\}$ satisfy $w(a + m) = w(a)$ for every $a \in \mathbb{Z}$. The period m is prescribed and need not be the least period. Define

$$S_w = \{a + w(a) : a \in \mathbb{Z}\}.$$

Every element $a + w(a)$ has minimal polynomial

$$x^2 - 2ax + a^2 + 1.$$

Thus distinct real parts lie in distinct minimal-polynomial classes, and every such fibre is a noncentral singleton. Nevertheless, many of the resulting infinite sets are ringsets.

Our main theorem is the following local–global classification.

Theorem 1.1. *Let $m \geq 1$ be odd and squarefree, and let $w : \mathbb{Z} \to \{\mathbf{i}, \mathbf{j}\}$ have period m. Then S_w is a ringset of $\mathbf{L}$ if and only if both of the following hold:*

1. *w is nonconstant;*
2. *for every prime $p \mid m$ with $p \equiv 3 \pmod 4$, and every $c \in \mathbb{Z}/p\mathbb{Z}$, the set*

$$\{w(a) : a \equiv c \pmod p\}$$

contains both $\mathbf{i}$ and $\mathbf{j}$.

The criterion is independent of whether the displayed period is minimal. At the split primes, no two-colour condition is needed.

Corollary 1.2. *Suppose that m is odd and squarefree and every prime divisor of m is congruent to 1 modulo 4. Then every nonconstant word of period m produces a ringset S_w. In particular, among the 2^m labelled binary words on $\mathbb{Z}/m\mathbb{Z}$, exactly $2^m - 2$ satisfy the conclusion.*

The finite-field core-set problem for $M_2(\mathbb{F}_q)$ has already been classified algorithmically by Werner [3]. Rissner and Werner [2] enumerated core subsets within each similarity class, counted purely core subsets globally, and obtained an asymptotic count for all core subsets. We do not claim those local classifications and counts, the split quaternion representation, or the residue-null-ideal bridge as new. The contribution here is the explicit split prime-power lemma and its combination with periodic congruence orbits and nonsplit obstructions to obtain the above infinite-family if-and-only-if theorem.

2 Null ideals and the residue criterion

Let A be a ring and $T \subseteq A$. Its null ideal is

$$\mathcal{N}(T, A) = \{f \in A[x] : f(t) = 0 \text{ for every } t \in T\}.$$

It is always a left ideal. The set T is called *core* if $\mathcal{N}(T, A)$ is a two-sided ideal. For $n \geq 2$, write $\overline{S}_n$ for the image of $S \subseteq \mathbf{L}$ in $\mathbf{L}/n\mathbf{L}$.

We use the following consequence of Werner [4, Lemmas 5.3 and 5.4].

Lemma 2.1 (residue criterion)**.** *Let $S \subseteq \mathbf{L}$.*

1. *If $\overline{S}_n$ is core in $\mathbf{L}/n\mathbf{L}$ for every $n \geq 2$, then S is a ringset.*
2. *Conversely, if for some n there are $g \in \mathcal{N}(\overline{S}_n, \mathbf{L}/n\mathbf{L})$ and $u \in \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ such that $gu \notin \mathcal{N}(\overline{S}_n, \mathbf{L}/n\mathbf{L})$, then S is not a ringset.*

Proof. Every $f \in \mathbb{D}[x]$ can be written as $f = g/n$ with $g \in \mathbf{L}[x]$ and $n \geq 1$. By [4, Lemma 5.3], f is integer-valued on S exactly when the reduction of g belongs to $\mathcal{N}(\overline{S}_n, \mathbf{L}/n\mathbf{L})$.

If every residue image is core, reduction remains null after the coefficients of g are right-multiplied by $\mathbf{i}, \mathbf{j}$, or $\mathbf{k}$. Hence $f\mathbf{i}, f\mathbf{j}, f\mathbf{k}$ are integer-valued on S. Since every Lipschitz quaternion is an integral linear combination of $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$, [4, Lemma 5.4] gives multiplicative closure. The same argument in reverse proves part 2 after lifting the coefficients of the residue polynomial g to $\mathbf{L}[x]$. □

The next elementary lemma records the paired-fibre calculation contained in the proof of [4, Proposition 5.6]. We state it directly for a finite quotient; we are not applying the global proposition itself to a finite ring.

Lemma 2.2 (two letters over every real residue)**.** *Let $q \geq 2$, let $A = \mathbf{L}/q\mathbf{L}$, and let*

$$T = \{a + \mathbf{i}, a + \mathbf{j} : a \in \mathbb{Z}/q\mathbb{Z}\}.$$

Then T is core in A.

Proof. Take $f \in \mathcal{N}(T, A)$ and fix a real residue a. Divide f by the central monic polynomial

$$\mu_a(x) = x^2 - 2ax + a^2 + 1.$$

The remainder has the form $r(x) = \gamma_1 x + \gamma_0$, and it vanishes at $a + \mathbf{i}$ and $a + \mathbf{j}$. Subtracting gives

$$\gamma_1(\mathbf{i} - \mathbf{j}) = 0.$$

Since

$$(\mathbf{i} - \mathbf{j})(-\mathbf{i} + \mathbf{j}) = 2,$$

we have $2\gamma_1 = 0$ in A; no invertibility of 2 is used.

For $u \in \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and $\alpha \in \{a + \mathbf{i}, a + \mathbf{j}\}$, direct right evaluation gives

$$(ru)(\alpha) - r(\alpha)u = \gamma_1(u\alpha - \alpha u).$$

Every displayed commutator lies in $2A$, so the right-hand side is zero. To pass from the remainder back to f, write $f = q_a\mu_a + r$. Since μ_a has central coefficients,

$$fu = (q_a u)\mu_a + ru, \qquad ((q_a u)\mu_a)(\alpha) = 0.$$

Hence $(fu)(\alpha) = (ru)(\alpha)$ at both roots of μ_a, and fu is null on T for the three basis units. It follows by linearity that the null ideal is stable under right multiplication by every constant in A. If $h(x) = \sum_t r_t x^t \in A[x]$, then

$$fh = \sum_t (fr_t)x^t = \sum_t x^t(fr_t)$$

belongs to the null ideal because the latter is a left ideal and x is central. Hence the null ideal is two-sided. □

The same proof applies to any subset of the two-letter grid that contains both letters above every real residue.

3 A split prime-power lemma

At a prime congruent to 1 modulo 4, the conclusion is much stronger: one letter above each real residue is enough.

Lemma 3.1 (split one-letter lemma)**.** *Let $p \equiv 1 \pmod 4$ be prime, let $e \geq 1$, and put $R = \mathbb{Z}/p^e\mathbb{Z}$ and $A = \mathbf{L}/p^e\mathbf{L}$. Suppose*

$$T \subseteq \{a + \mathbf{i}, a + \mathbf{j} : a \in R\}$$

contains at least one point of real part a for every $a \in R$. Then T is core in A.

Proof. Hensel lifting gives a unit $s \in R$ with $s^2 = -1$. The assignments

$$\mathbf{i} \longmapsto \begin{pmatrix} s & 0 \\ 0 & -s \end{pmatrix}, \qquad \mathbf{j} \longmapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

give an isomorphism $A \simeq M_2(R)$. Indeed, $a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ maps to

$$\begin{pmatrix} a + bs & c + ds \\ -c + ds & a - bs \end{pmatrix},$$

and the four coordinates are recovered by dividing sums and differences by the units 2 and s.

For $u \in \{\mathbf{i}, \mathbf{j}\}$ define

$$e_u^+ = \frac{1+u/s}{2}, \qquad e_u^- = \frac{1-u/s}{2}.$$

These are complementary orthogonal idempotents. For every $a \in R$ and $d \geq 0$,

$$(a+u)^d = (a+s)^d e_u^+ + (a-s)^d e_u^-.$$

Under the matrix representation, their image modules are

$$\begin{aligned} \operatorname{im}(e_{\mathbf{i}}^+) &= R(1,0)^T, \quad \operatorname{im}(e_{\mathbf{i}}^-) = R(0,1)^T, \\ \operatorname{im}(e_{\mathbf{j}}^+) &= R(1,s)^T, \quad \operatorname{im}(e_{\mathbf{j}}^-) = R(1,-s)^T. \end{aligned}$$

For $u_1, u_2 \in \{\mathbf{i}, \mathbf{j}\}$, place a generator of $\operatorname{im}(e_{u_1}^+)$ beside a generator of $\operatorname{im}(e_{u_2}^-)$. In the four cases

$$(u_1, u_2) = (\mathbf{i}, \mathbf{i}), (\mathbf{i}, \mathbf{j}), (\mathbf{j}, \mathbf{i}), (\mathbf{j}, \mathbf{j}),$$

the determinants may be chosen as

$$1, \qquad -s, \qquad 1, \qquad -2s,$$

respectively. They are all units. Thus the two image modules are complementary; if a matrix X satisfies $Xe_{u_1}^+ = Xe_{u_2}^- = 0$, then $X = 0$.

Now take $f(x) = \sum_d c_d x^d \in \mathcal{N}(T, A)$ and introduce the scalar-variable matrix polynomial

$$F(z) = \sum_d c_d z^d, \qquad z \in R.$$

For a fixed z, use the selected point of real part $z-s$ and multiply its zero-evaluation equation on the right by its plus idempotent. This gives

$$F(z)e_{u_1}^+ = 0.$$

Similarly, the selected point of real part $z+s$ gives

$$F(z)e_{u_2}^- = 0.$$

The complementarity just proved implies $F(z) = 0$ for every $z \in R$. This argument does not use uniqueness of polynomial functions over R.

Let $r \in A$ and let fr denote uniform right multiplication of the coefficients by r. Its scalar-variable polynomial is

$$F_r(z) = \sum_d (c_d r) z^d = F(z)r = 0.$$

Applying the idempotent decomposition again shows that fr vanishes at every point of T. Hence the null ideal is stable under right multiplication by all constants. As in Lemma 2.2, stability under arbitrary polynomial right multiplication follows from the left-ideal property and centrality of x. Therefore T is core. □

The proof deliberately uses the matrix equations $Xe = 0$ and the image modules of the idempotents. No classification of left or right ideals is being invoked.

4 Periodic congruence fibres

The arithmetic input is elementary but records exactly where squarefreeness enters.

Lemma 4.1 (periodic orbit lemma)**.** *Let w have period m, let $q \geq 1$, and fix $a \in \mathbb{Z}/q\mathbb{Z}$. The positions modulo m attained by integers congruent to a modulo q are*

$$a + q\mathbb{Z} \pmod{m} = \{r \in \mathbb{Z}/m\mathbb{Z} : r \equiv a \pmod{\gcd(q, m)}\}.$$

Proof. The subgroup generated by q in $\mathbb{Z}/m\mathbb{Z}$ consists exactly of the multiples of $\gcd(q, m)$. Translating by a gives the formula. □

In particular, if $\gcd(q, m) = 1$, every real residue fibre of S_w contains both letters exactly when w is nonconstant. If m is squarefree, $p \mid m$, and $q = p^e$, then $\gcd(q, m) = p$, so a real residue fibre sees exactly one position class modulo p.

The formula also shows that the criterion does not depend on choosing a nonminimal period. If the least period is $d \mid m$ and a prime divisor of m does not divide d, each corresponding position class still projects onto every position modulo d.

5 Nonsplit missing-letter obstructions

We next give the explicit obstruction used for necessity.

Lemma 5.1 (missing-letter obstruction)**.** *Let $p \equiv 3 \pmod{4}$ be prime and let*

$$T \subseteq \{a + \mathbf{i}, a + \mathbf{j} : a \in \mathbb{F}_p\}.$$

If for some a the fibre of real part a is nonempty and contains only one of the two letters, then T is not core in $\mathbf{L}/p\mathbf{L}$.

Proof. In $(\mathbf{L}/p\mathbf{L})[x]$ define the central polynomial

$$\rho_p(x) = \frac{x + x^p}{2}.$$

For $y = b + u$ with $u \in \{\mathbf{i}, \mathbf{j}\}$, the congruence $p \equiv 3 \pmod{4}$ gives $u^p = -u$, and hence

$$\rho_p(y) = b.$$

The central Lagrange indicator of the real residue a is

$$\delta_a(x) = 1 - (\rho_p(x) - a)^{p-1}.$$

It evaluates to 1 above a and to 0 above every other real residue.

Suppose first that the fibre contains $a + \mathbf{i}$ but not $a + \mathbf{j}$. Put

$$h(x) = (x - (a + \mathbf{i}))\delta_a(x).$$

The order of the two factors is intentional. Since δ_a has central coefficients, h vanishes on all of T. After uniformly right-multiplying its coefficients by $\mathbf{j}$, evaluation at $\alpha = a + \mathbf{i}$ gives

$$(h\mathbf{j})(\alpha) = \mathbf{j}\alpha - \alpha\mathbf{j} = \mathbf{ji} - \mathbf{ij} = -2\mathbf{k} \neq 0.$$

Thus the null ideal is not stable under right multiplication. If the fibre contains only $a + \mathbf{j}$, use $(x - (a + \mathbf{j}))\delta_a(x)$ and right multiplication by $\mathbf{i}$; the value is $2\mathbf{k} \neq 0$. □

Choosing integer representatives of the coefficients lifts the above null polynomial to $\widehat{h} \in \mathbf{L}[x]$. Then $\widehat{h}/p$ is integer-valued on every integer set whose residue image is T, while the indicated right multiple is not. Lemma 2.1 therefore turns the finite obstruction into a non-ringset certificate.

6 Proof of the classification

We now prove Theorem 1.1.

Sufficiency. Assume that w is nonconstant and that every position class required in condition 2 is bichromatic. We show that the image of S_w is core modulo every prime power p^e.

If $p \equiv 1 \pmod 4$, every real residue modulo p^e is represented by at least one element of S_w. Lemma 3.1 applies without any two-colour condition.

Suppose $p = 2$. Since m is odd, $\gcd(2^e, m) = 1$. Lemma 4.1 and nonconstancy show that every real residue fibre modulo 2^e contains both letters. Lemma 2.2 applies.

Now let $p \equiv 3 \pmod 4$. If $p \nmid m$, then again $\gcd(p^e, m) = 1$, and Lemmas 4.1 and 2.2 apply. If $p \mid m$, squarefreeness gives $\gcd(p^e, m) = p$. Each real residue fibre therefore sees exactly one position class modulo p, which is bichromatic by condition 2. Lemma 2.2 again applies.

It remains to pass from prime powers to arbitrary denominators. Write $n = \prod_r p_r^{e_r}$. The Chinese remainder theorem gives

$$\mathbf{L}/n\mathbf{L} \simeq \prod_r \mathbf{L}/p_r^{e_r}\mathbf{L}.$$

For $A = \prod_r A_r$ and an arbitrary subset $T \subseteq A$, polynomial coefficients and right evaluation are componentwise. Therefore

$$\mathcal{N}(T,A) \simeq \prod_r \mathcal{N}(\pi_r T, A_r).$$

This identity does not require T to be the Cartesian product of its projections: a component polynomial vanishes on every element of T exactly when it vanishes on the corresponding projection. Since every prime-power projection of S_w is core, its image modulo n is core. Lemma 2.1 now shows that S_w is a ringset.

Necessity. If w is constant, reduce modulo 3. Every real residue fibre then contains only one letter, so Lemma 5.1 and Lemma 2.1 show that S_w is not a ringset.

Next suppose that $p \mid m$, $p \equiv 3 \pmod 4$, and some position class $c \pmod p$ is monochromatic. Fix a real residue $a \equiv c \pmod p$. By Lemma 4.1, the fibre of the image of S_w above a sees precisely that position class, and hence contains only one letter. Lemma 5.1 again gives a null polynomial whose right multiple is non-null. Lemma 2.1 implies that S_w is not a ringset. This proves both necessary conditions and completes the theorem.

7 Examples and the squarefree boundary

If $m = 5$, the only prime divisor is split. Hence every one of the $2^5 - 2 = 30$ nonconstant labelled words produces a ringset. More generally, Corollary 1.2 gives $2^m - 2$ examples whenever all prime divisors of m are 1 modulo 4.

For $m = 3$ or $m = 7$, condition 2 is impossible: a position class modulo m contains only one position. Thus no word of these prescribed periods gives a ringset. Composite periods can behave differently. For $m = 15$, the condition requires each of the three position classes modulo 3 to be bichromatic. For $m = 21$, both every class modulo 3 and every class modulo 7 must be bichromatic. The length-m word obtained by alternating the two letters on the standard representatives $0, 1, \ldots, m-1$ satisfies these requirements for $m = 15, 21$, and similarly for $m = 35$.

Squarefreeness is used only in the equality $\gcd(p^e, m) = p$ when a nonsplit prime divides m. If $p^2 \mid m$, a fibre modulo p^e may see a position class modulo p^2 or a higher power. Being bichromatic modulo p no longer guarantees a paired fine fibre, while a missing letter in one fine fibre is not automatically detected by the finite-field polynomial of Lemma 5.1. A classification for nonsquarefree periods therefore requires genuinely local-ring information about core subsets of $\mathbf{L}/p^e\mathbf{L}$. We make no claim about that boundary here.

8 Computational calibration

The proof above is symbolic and does not depend on computation. During discovery, the degree-$\leq 2p$ truncated evaluation kernel over $\mathbb{F}_p$ was checked for all three choices `i only`, `j only`, or `both` over every real residue:

- all $3^3 = 27$ patterns for $p = 3$;
- all $3^5 = 243$ patterns for $p = 5$;
- all $3^7 = 2187$ patterns for $p = 7$;

- representative patterns for $p = 11$ and $p = 13$.

Within this degree-$\leq 2p$ truncation, for $p \equiv 3 \pmod 4$ only the all-bichromatic pattern was stable; for $p \equiv 1 \pmod 4$, every pattern was stable. These checks motivated the local lemmas but do not prove them or establish stability of the full null ideal. The discovery program is retained with the research record and explicitly labels itself as finite-field calibration.

AI assistance was used to help formulate finite test cases, review the exploratory program, and compare its reported outputs with the symbolic argument. The author executed and inspected the computations; no computed case is used as a proof of the theorem.

9 Prior-work boundary

The following components are prior work and are not claimed here:

- the general connection between ringsets and null-ideal sets [1];
- the algorithmic classification of finite core subsets of $M_2(\mathbb{F}_q)$ [3], and the within-class, purely-core, and asymptotic counts in [2];
- the finite Lipschitz/Hurwitz ringset classification, the residue criterion, the right-multiplication criterion, and the infinite sufficient condition [4].

As of the literature search completed on 23 August 2026, we found no public source directly stating the odd-squarefree periodic criterion of Theorem 1.1. This search result is a bounded prior-work assessment, not a claim of absolute global priority. The result has not yet received external expert review or formal peer review.

Declaration of generative AI and AI-assisted technologies

During the research and preparation of this work, the author used OpenAI Codex to assist with literature searches, symbolic exploration, exploratory code review, proof checking, and language drafting. The author reviewed and edited all generated output and takes full responsibility for the content of this manuscript.

References

[1] Sophie Frisch, *Polynomial functions on subsets of non-commutative rings—a link between ringsets and null-ideal sets*, ITM Web of Conferences **20** (2018), 01003. doi:10.1051/itmconf/20182001003.

[2] Roswitha Rissner and Nicholas J. Werner, *Counting core sets in matrix rings over finite fields*, Linear Algebra and its Applications **720** (2025), 1–25. doi:10.1016/j.laa.2025.04.006; arXiv:2405.04106v2.

[3] Nicholas J. Werner, *Null ideals of subsets of matrix rings over fields*, Linear Algebra and its Applications **642** (2022), 50–72. doi:10.1016/j.laa.2022.02.007.

[4] Nicholas J. Werner, *Integer-valued polynomials on subsets of quaternion algebras*, Journal of Algebra **686** (2026), 195–219. doi:10.1016/j.jalgebra.2025.08.011; arXiv:2412.20609v2.